

Stenac Effervescent Tablets 600 mg (Acetylcysteine)

COMPOSITION

Each tablet contains :

Acetylcysteine600 mg

DESCRIPTION

White to light yellow, round, flat-faced tablets, embossed with "SYN" on one side.

INDICATIONS

Reduce the viscosity of the secretion of mucous membrane of respiratory tract.

Antidotes for acute poisoning: Acetaminophen overdose.

ACTIONS (PHARMACODYNAMIC)

Acetylcysteine is the main active composition in the effervescent tablets, through directly interaction with mucoprotein complex and nucleic acid then to disrupt the mucoprotein of phlegm into smaller clear units and thereby reduces the viscosity of the phlegm. Acetylcysteine as a mucolytic agent results from its sulfhydryl interacting with disulfide bonds in mucoprotein, with the mucus subsequently being broken into less viscous units.

Acetylcysteine, an antioxidant, with active free sulphahydryl group can interact with reactive oxygen species. Acetylcysteine have a therapeutic application in preventing oxidative tissue injury induced by hypochlorous acid (HOCl). HOCl is toxic to cells because it initiates oxidative injury and causes the activation of phagocyte peroxidases, such as the neutrophil enzyme MPO. Acetylcysteine has been shown to dramatically reduce the viscosity and tenacity of sputum. The liquefying action is due to the presence of a free sulphhydryl group which opens up disulphide bonds present in mucus. Pretreatment with Acetylcysteine may prevent lung injury by diminishing elastase activity; this diminished activity of elastase may be achieved by enhanced inactivation by antiproteases.

It is concluded that acetylcysteine is useful for viscosity and maturative secretion of phlegm induced by acute and chronic respiratory system inflammation. In addition, the structure of acetylcysteine can transport through cell membrane and decompose to L-cysteine, the main component of glutathione (GSH), and maintains cellular GSH at a level. GSH, an antioxidant, protects cells from toxins such as free radicals.

Acetaminophen overdose:

The primary toxic effect of acetaminophen is hepatotoxicity caused by the formation of the toxic metabolite N-acetyl-pbenzoquinonimine (NAPQI). NAPQI is conjugated by glutathione at therapeutic dosages. In an overdose, the normal pathways of metabolism are overwhelmed and NAPQI is formed in greater quantities, resulting in depletion of glutathione and hepatic injury. Administration of NAC is beneficial in preventing or mitigating hepatic injury through stimulation of glutathione synthesis, enhancing nontoxic routes of acetaminophen metabolism, detoxifying the toxic metabolite and free radical scavenging.

PHARMACOKINETICS

Acetylcysteine is rapidly absorbed from the gastrointestinal tract and peak plasma concentrations occur about 0.5 to 1 hour after oral doses of 200 to 600mg. Some studies indicate dose-dependent pharmacokinetics with peak concentrations, the time taken to reach peak concentrations, and bioavailability increasing with increasing doses.

Acetylcysteine may be present in plasma as the parent compound or as various oxidized metabolites such as N-acetylcystine, N,N-diacetylcystine, and cysteine either free or bound to plasma proteins by labile disulfide bonds or as a fraction incorporated into protein peptide chains. In a study about 50% was in a covalently protein-bound form 4 hours after a dose. Oral bioavailability is low and means values have ranged from 4 to 10% depending on whether total acetylcysteine or just the reduced forms are measured. It has been suggested that acetylcysteine's low oral bioavailability may be due to metabolism in the gut wall and first-pass metabolism in the liver. Renal clearance may account for about 30% of total body clearance. On intravenous dosage mean terminal half-lives have been calculated to be 1.95 and 5.58 hours for reduced and total acetylcysteine, respectively; the terminal half-life of total acetylcysteine was 6.25 hours after oral doses. Absorption of NAC is rapid following oral administration, but the bioavailability is only 6-10% due to extensive firstpass metabolism. In addition, oral administration is often limited by nausea and vomiting which results in delayed or ineffective administration of NAC. Intravenous administration of NAC results in 100% bioavailability. The mean terminal half-life is approximately 6 hours. Clearance is not reduced and plasma concentrations are not increased in patients with severe liver damage. Currently there is no dosage adjustment recommended for patients with renal or hepatic insufficiency.

DOSAGE AND ADMINISTRATION

Adults: Take one effervescent tablet once a day. Dissolve the tablet in a glass with some water and mix it. It is better to take at night, avoiding the addition of drugs to the solution and taking it before or after meals.

Acetaminophen poisoning: Children and Adults

140mg/kg; followed by 17 doses of 70mg/kg every 4 hour; repeat dose if emesis occurs within 1 hour of administration.

For oral use: Keep medicine out of reach of children.

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CONTRAINDICATIONS

Contraindicated in children below two (2) years of age.

It is contraindicated in patients with hypersensitivity to acetylcysteine, or any of the excipients. As Stenac Effervescent Tablets 600 mg contains aspartame, it is contraindicated in patients suffering from phenylketonuria.

INTERACTION WITH OTHER MEDICAL PRODUCTS AND OTHER FORMS OF INTERACTION

Stenac Effervescent Tablets 600 mg can be administered concurrently with amoxycillin, doxycillin and erythromycin. When other oral antibiotics or drugs are required, they should be administered 1-2 hours apart from Stenac Effervescent Tablets 600 mg.

PREGNANCY AND LACTATION

Category B: Either animal-reproduction studies have not demonstrated a foetal risk but there are no controlled studies in pregnant women or animal-reproduction studies have shown an adverse effect (other than a decrease in fertility) that was not confirmed in controlled studies in women in the 1st trimester (and there is no evidence of a risk in later trimesters). The administration of acetylcysteine during pregnancy is advised only if there are compelling reasons.

SYMPTOMS AND TREATMENT OF OVERDOSE

There is no specific antidote for acetylcysteine and treatment is symptomatic. It consists of postural drainage. Bronchial suction and supporting therapy as indicated.

ADVERSE REACTIONS

Nausea, heartburn, vomiting, urticaria, headache and tinnitus have infrequently been reported but these rarely necessitate withdrawal of treatment. Allergic skin reactions and bronchospasm, especially in asthmatic patients, have been reported rarely in patients taking oral acetylcysteine. High doses of acetylcysteine given intravenously have caused anaphylactoid reactions.

Immune System Disorders:

Anaphylactic / anaphylactoid reaction

Skin and Subcutaneous Tissue Disorders:

Severe cutaneous adverse reactions (SCAR) e.g. erythema multiforme, Stevens-Johnson syndrome(SJS) and toxic epidermal necrolysis (TEN). In most of these cases reported at least one other drug was administered at the same time, which may have possibly enhanced the described mucocutaneous effects.

PRECAUTIONS

Patients with asthma should be carefully monitored. Treatment with acetylcysteine should be stopped when bronchospasm occurred.

The sulfur odour on opening the effervescent tablet does not indicate deterioration of the product, but is due to the active ingredient.

STORAGE

30 tablets are available as packs of aluminium foil in a carton box.

Store below 30° C.

SHELF LIFE:

Three years

REG. NO.: MAL09011475X

MANUFACTURER:

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